Structures Coursework

Group 8

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Contents

1 Span calculation **2**
- 1.1 Required plastic moment for span AB . . . 2
- 1.2 Plastic analysis for span CD . . . 3
- 1.3 Plastic analysis of span BC . . . 4

2 Deliverable 1: Geometry **6**

3 Deliverable 2: Material, Section, and Loading Data **7**
- 3.1 Material and Section Properties and Loading Data . . . 7

4 Deliverable 3: Initial Model Validation **8**
- 4.1 Assumptions . . . 8
- 4.2 Model Validation . . . 8
- 4.3 Model Adequacy . . . 8

5 Deliverable 4: Mesh Refinement Sensitivity Study **9**
- 5.1 Convergence Study Methodology . . . 9
- 5.2 Conclusion . . . 9

6 Deliverable 5: Influence of Prestressing Assumptions **10**
- 6.1 Quantitative Comparison . . . 10
- 6.2 Discussion of Results . . . 10
- 6.3 Implications for Design and Tolerance . . . 10

7 Deliverable 6: Boundary Conditions for More Detailed Analysis **11**

1 Span calculation

1.1 Required plastic moment for span AB

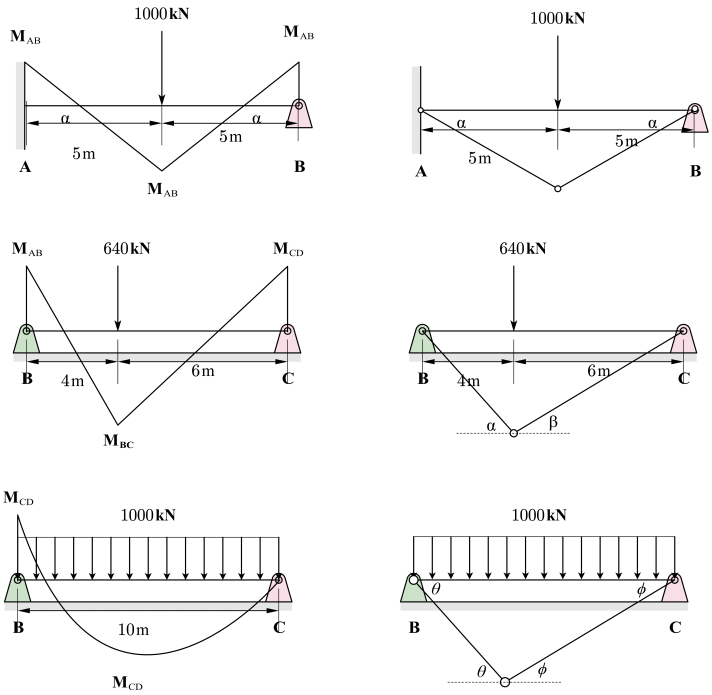


Figure 1: The three sections of the whole span

Span $AB = 10\,\mathrm{m}$, with $P = 1000\,\mathrm{kN}$ acting at midspan. Let the end rotations be θ. Then the deflection at the load point is

$$\delta = 5\theta$$

The plastic hinge rotation at the load point is

$$\theta + \theta = 2\theta$$

The internal work is therefore

$$W_i = M_{p,AB}(\theta + 2\theta + \theta) = 4M_{p,AB}\theta$$

The external work is

$$W_e = 1000(5\theta) = 5000\theta$$

Equating external and internal work,

$$5000\theta = 4M_{p,AB}\theta$$

Hence,

$$M_{p,AB} = 1250\,\mathrm{kNm}$$

1.2 Plastic analysis for span CD

Compatibility at the hinge

From compatibility,

$$x\theta = (12 - x)\phi$$

so that

$$\phi = \frac{x}{12 - x}\theta$$

External work

The deflected shape is triangular on each side, so

$$W_e = w \times \text{area under deflection shape}$$

Thus,

$$W_e = 100\left(\frac{1}{2} \cdot 12 \cdot x\theta\right) = 600x\theta$$

Internal work

At C, the hinge rotation is

$$\theta$$

At the positive hinge, the rotation is

$$\theta + \phi = \theta + \frac{x}{12 - x}\theta = \theta\left(\frac{12}{12 - x}\right)$$

Hence,

$$W_i = M_{p,CD}\left[\theta + \theta\left(\frac{12}{12 - x}\right)\right] = M_{p,CD}\theta\left(\frac{24 - x}{12 - x}\right)$$

Equating external and internal work gives

$$600x\theta = M_{p,CD}\theta\left(\frac{24 - x}{12 - x}\right)$$

Therefore,

$$M_{p,CD} = 600x\left(\frac{12 - x}{24 - x}\right)$$

Critical mechanism

To obtain the critical mechanism, maximize $M_{p,CD}$ with respect to x:

$$\frac{dM_{p,CD}}{dx} = 0$$

which gives

$$x = 12(2 - \sqrt{2}) = 7.03\,\text{m}$$

Hence,

$$M_{p,CD} = 600(7.03)\left(\frac{12 - 7.03}{24 - 7.03}\right)$$

Therefore,

$$M_{p,CD} \approx 1235\,\text{kNm}$$

1.3 Plastic analysis of span BC

Assume that the end plastic moments are known as

$$M_{AB} = 1250\,\text{kNm}, \qquad M_{CD} = 1235\,\text{kNm}$$

and that an additional plastic hinge forms within span BC under the applied load $P = 640\,\text{kN}$. The span length is

$$BC = 4 + 6 = 10\,\text{m}$$

with the point load located 4 m from support B and 6 m from support C.

Collapse mechanism

Let the rotation at end B be θ_B, and the rotation at end C be θ_C. Since the vertical displacement of the load point must be the same whether measured from the left or the right segment,

$$4\theta_B = 6\theta_C$$

Hence,

$$\theta_C = \frac{4}{6}\theta_B = \frac{2}{3}\theta_B$$

The vertical displacement of the load point is therefore

$$\delta = 4\theta_B$$

External work

The external work done by the applied load is

$$W_e = P\delta = 640(4\theta_B) = 2560\theta_B$$

Internal work

Internal work is done at the three plastic hinges:

- at B: rotation $= \theta_B$, moment $= M_{AB}$
- at the internal hinge in span BC: rotation $= \theta_B + \theta_C$, moment $= M_{BC}$
- at C: rotation $= \theta_C$, moment $= M_{CD}$

Thus,

$$W_i = M_{AB}\theta_B + M_{BC}(\theta_B + \theta_C) + M_{CD}\theta_C$$

Substituting the known values and $\theta_C = \frac{2}{3}\theta_B$,

$$W_i = 1250\theta_B + M_{BC}\left(\theta_B + \frac{2}{3}\theta_B\right) + 1235\left(\frac{2}{3}\theta_B\right)$$

$$W_i = 1250\theta_B + \frac{5}{3}M_{BC}\theta_B + 823.33\theta_B$$

$$W_i = \left(2073.33 + \frac{5}{3}M_{BC}\right)\theta_B$$

Equating external and internal work

At collapse,

$$W_e = W_i$$

so

$$2560\theta_B = \left(2073.33 + \frac{5}{3}M_{BC}\right)\theta_B$$

Cancelling θ_B,

$$2560 = 2073.33 + \frac{5}{3}M_{BC}$$

$$\frac{5}{3}M_{BC} = 486.67$$

$$M_{BC} = 486.67 \times \frac{3}{5}$$

$$M_{BC} \approx 292\,\text{kNm}$$

Result

Therefore, the required plastic moment capacity of the internal hinge in span BC is

$$M_{BC} \approx 292\,\text{kNm}$$

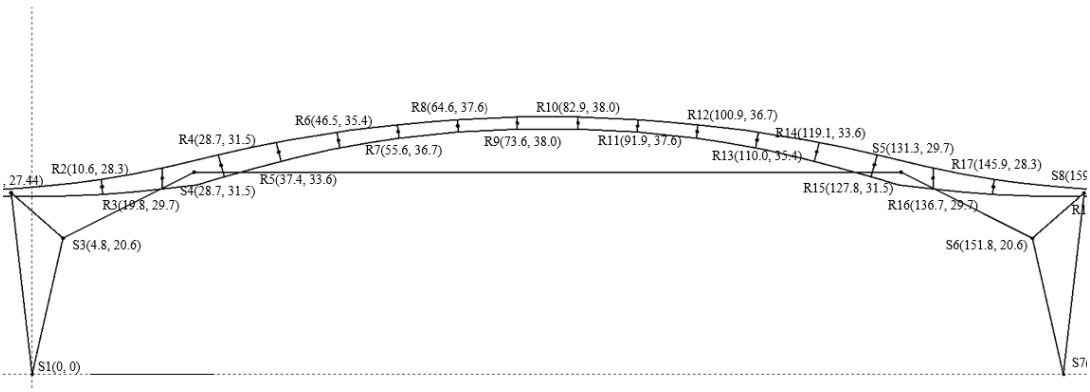
Based on the above result, the plastic moment required at the assumed internal hinge in span BC is lower than the corresponding plastic moment capacities associated with the end sections. If a single fully plastic moment capacity M_p is assumed for both sagging and hogging bending, this implies that the section in span BC would reach the fully plastic state earlier than the end sections B and C. Consequently, the assumed three-hinge collapse mechanism is not kinematically admissible as the governing failure mechanism for the beam, because the plastic hinge in span BC would form before the other required hinges had fully developed. In other words, under the assumption that the positive and negative plastic moment capacities are equal, span BC would not sustain the assumed failure mechanism as drawn; instead, yielding would initiate there prematurely and the actual collapse mechanism would need to be reassessed on the basis of the weaker section controlling the plastic redistribution process.

2 Deliverable 1: Geometry

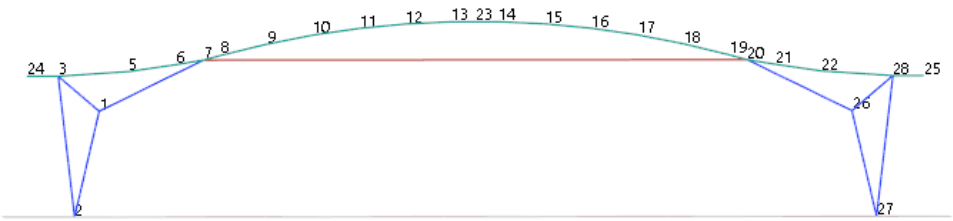
The roof geometry was reconstructed from the supplementary drawing using a one-to-one SketchUp model and then transferred into GSA. As shown in Figure 2(a), points labelled R define the roof profile and represent the mid-points of the roof section, while points labelled S define the support system and lower restraint points.

The second diagram shows the FE idealisation adopted for analysis. In Figure 2(b), green denotes beam elements used for the roof profile, red denotes the high tie modelled as a tie element, and blue denotes the support-tree members modelled as bar elements. In GSA, beam elements model axial, bending and torsional effects, bar elements provide axial stiffness only, and tie elements are axial-only elements with zero stiffness in compression. This makes beams suitable for the roof rafters, bars suitable for the tree members, and a tie suitable for the prestressed high tie. Both end supports were modelled as fixed supports, while the pin-connected members were treated consistently on both sides because the connected elements were assumed not to transfer bending moment. This idealisation captures the main load path and preserves the key dimensions required for later loading, validation and mesh sensitivity studies.

Because both the arch rafter and tusk rafter vary in depth along their length, the roof-profile members were implemented in GSA using tapered 1D beam elements rather than prismatic sections. Oasys GSA states that a 1D element can taper between end 1 and end 2 when it is assigned a tapering section profile with two section sizes, and that when a tapering member is divided into multiple elements a taper offset is used so that each element retains the correct local dimensions from its position along the original member [2]. This is important for the present roof model because the varying rafter depth changes the member bending stiffness along the span, so representing that taper improves the fidelity of the global frame idealisation.



(a) Reconstructed nodal geometry



(b) Idealised FE member types

Figure 2: Geometry adopted for the global model.

3 Deliverable 2: Material, Section, and Loading Data

3.1 Material and Section Properties and Loading Data

The global FE model was defined using linear-elastic steel. In accordance with the brief, all members were assigned an elastic modulus of 200 GPa, a Poisson's ratio of 0.30 and density of 7850 kg m^{-3} were adopted so that self-weight could be included automatically in GSA. S235 steel was adopted for the primary roof members because its elastic modulus complies with the brief requirement, while other material parameters such as shear modulus and strength are not governing in this linear-elastic global analysis. The twin high-tie strands were modelled as a single prestressed tie element with the same total cross-sectional area and an initial prestress of 450 MPa.

Only vertical symmetric loading was considered. The characteristic permanent roof load was taken as the sum of the aluminium standing seam roof (0.10 kPa), structural deck (0.15 kPa), insulation/acoustic layers (0.40 kPa) and the service allowance (0.15 kPa), giving $G_k = 0.80$ kPa. The live load imposed on the roof was taken as $Q_k = 0.60$ kPa. The governing vertical ultimate limit state (ULS) combination used in the model was

$$1.35G_k + 1.5Q_k \tag{1}$$

Here, G_k and Q_k were converted to line loads using the projected tributary width carried by one transverse frame. The supplementary roof information states that the main rafters and associated tie-rods are spaced at 18 m centres, so a tributary width of 18 m was adopted for the 2D frame idealisation. On this basis, the superimposed permanent roof load is 14.4 kN m^{-1} (0.80×18) and the imposed live load is 10.8 kN m^{-1} (0.60×18), with member self-weight still included separately through the material density. This projected-area treatment is consistent with common roof-loading practice and with the basis commonly used for roof snow loading in Eurocode 1 [1].

The strap members were modelled as bar elements. Therefore, only their axial stiffness was represented in the global model, using an equivalent steel area of $A = 2 \times 500 \times 40 = 40\,000$ mm^2. The spacing between the two plates was neglected in this idealisation because the bar elements do not carry bending.

The adopted member sections and the associated modelling notes are summarised in Table 1.

Table 1: Section data adopted for the global model

Member	Section	Modelling Note
Arch rafter	Welded box, 800 mm wide, 50 mm wall thickness	Depth varies from about 1.8 m to 3.8 m along the roof profile
Tusk rafter	Welded I-section, 50 mm plate thickness	Overall depth varies from about 0.9 m to 3.8 m
Legs and arms	Circular Hollow Section (CHS) 914 mm × 40 mm	Taken directly from the supplementary sketch
Wings	CHS 711 mm × 50 mm	Taken directly from the supplementary sketch
High tie	Twin locked-coil strands, 2 × 115 mm	Modelled as one equivalent tie element with the same total cross-sectional area; prestressed to 450 MPa
Straps	Equivalent rod area 40 000 mm^2	Based on two 500 mm × 40 mm plates; bending stiffness neglected

4 Deliverable 3: Initial Model Validation

4.1 Assumptions

The validation assumes a 2D in-plane frame subjected to vertical symmetric loading. Out-of-plane bending, torsion, adjacent-frame load sharing and detailed connection stiffness are neglected, so the check is intended only to confirm that the global model gives a realistic level of response rather than an exact local solution.

4.2 Model Validation

As the prestressed arch-and-tie behaviour is difficult to analyse by simple hand calculations, the initial validation was carried out on the non-prestressed model and focused on the bending moment at the tusk-to-arch connection. The loaded tusk was idealised as a 25 m cantilever under the equivalent ULS line load $w = 1.35 \times 14.4 + 1.5 \times 10.8 = 35.64\,\mathrm{kN\,m^{-1}}$, giving a support moment $M = wL^2/2 \approx 11\,125\,\mathrm{kN\,m}$, as shown in Figure 3. The corresponding FE result is about 16 490 kN m, so the model predicts the correct order of magnitude for this key bending action. The difference is expected because the hand check neglects the stiffness of the curved arch, frame interaction between both sides of the roof, and the more complex redistribution captured by the FE model.

An additional source of modelling error was observed at the high-tie connection. The FE model predicts a relatively large vertical deformation there, even though the vertical nodal force associated with the high tie is small compared with the forces transferred by the surrounding members. Using the largest adjacent vertical force from the connection check, the high-tie vertical force is only about $114.6/5901 \approx 1.9\%$ of the neighbouring member force. This suggests that the large vertical movement is more likely to arise from compatibility within the simplified 2D global model and the axial-only idealisation of the tie than from a large vertical load carried by the high tie itself.

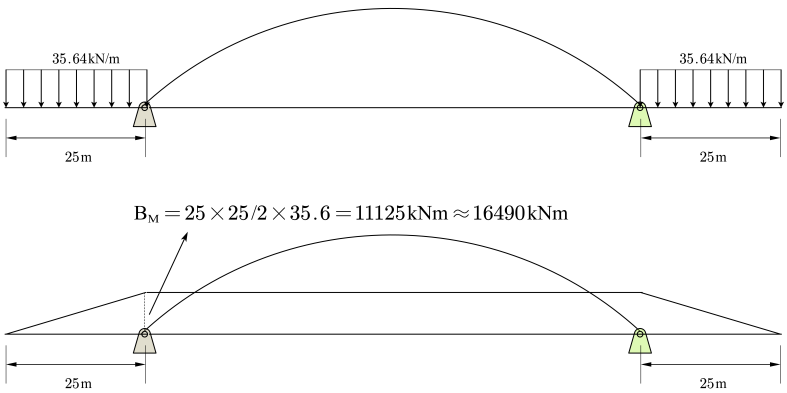


Figure 3: Hand validation of the tusk-to-arch connection bending moment.

4.3 Model Adequacy

Basic verification checks were also completed: the GSA solver ran without instability warnings, the vertical reactions balanced the applied gravity loads, and the reconstructed model matched the original drawing on both sides with an overall geometric deviation of approximately 0.5%. Taken together, the modelling assumptions, validation considerations and verification checks indicate that the global frame model is adequate for an initial study of the roof under symmetric vertical loading. However, it should still be understood as a simplified global model rather than a full representation of the three-dimensional structural behaviour.

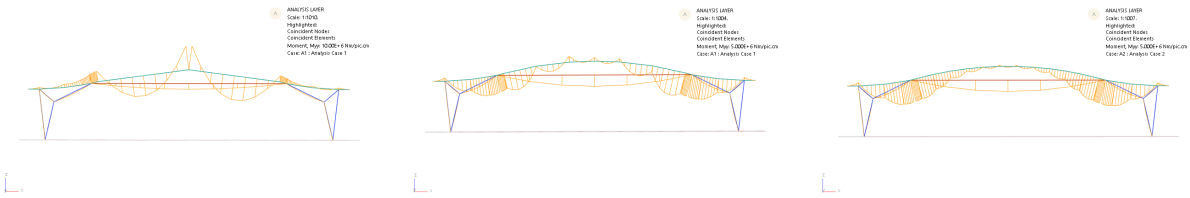
5 Deliverable 4: Mesh Refinement Sensitivity Study

5.1 Convergence Study Methodology

Three mesh densities were evaluated under the governing ULS vertical load combination $1.35G_k + 1.5Q_k + 1.0P_s$, where P_s represents the prestress in the high-tie. Mesh 1 is the coarsest representation of the arch, Mesh 2 retains more intermediate points, and Mesh 3 preserves the full coordinate set taken from the supplementary geometry; it is used as the reference mesh for comparison and is also the mesh adopted for the remainder of the report. To judge convergence, one displacement quantity and two force quantities were tracked: arch central deflection, high-tie axial force and arch central bending moment. These parameters give a useful picture of both overall stiffness and the redistribution of internal action as the curved geometry is refined.

Table 2: Influence of arch mesh density on key global responses

Mesh	Arch Elements	DoF	Arch Central Deflection (mm)	High-Tie Axial Force (kN)	Arch Central Bending Moment (kNm)	% Deflection Change vs Mesh 3
Mesh 1	14	53	241.2	9725	12760	37.6%
Mesh 2	16	59	175.7	9309	960.2	0.2%
Mesh 3	22	77	175.3	9320	198.2	0.0%



Mesh 1 Mesh 2 Mesh 3

Figure 4: Bending-moment diagrams for the three mesh densities used in the sensitivity study.

Table 2 and Figure 4 show that the coarsest arch discretisation significantly distorts the response. Relative to Mesh 3, Mesh 1 gives much larger deflection and a very different bending-moment pattern. Moving from Mesh 1 to Mesh 2, the sign of bending moment changes across parts of the arch, showing that the coarsest mesh does not reproduce the qualitative moment distribution reliably. This can be explained by the geometric idealisation: when the arch is represented by only a few straight segments, part of its curvature and stiffness is lost, so the model behaves more like the softer, unprestressed system. Mesh 2 is already close to Mesh 3 for deflection and tie force, but the central bending moment remains much more sensitive to curvature, so Mesh 3 is still required if bending results are to be used with confidence.

5.2 Conclusion

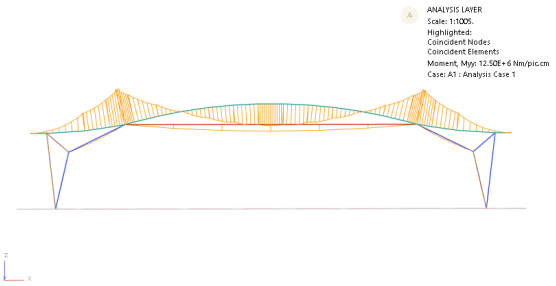
Based on these results, Mesh 3 is adopted for the remainder of the report. Although Mesh 2 is already adequate for deflection and tie force, only Mesh 3 captures the arch curvature accurately enough to stabilise the central bending moment, which is an important response quantity for interpreting the behaviour of the roof frame.

6 Deliverable 5: Influence of Prestressing Assumptions

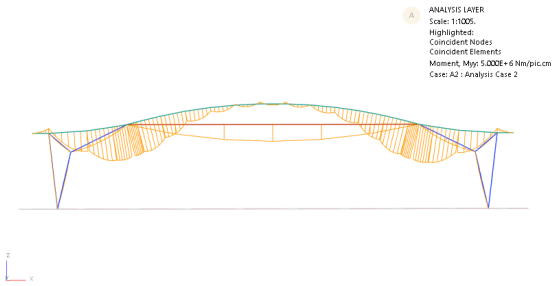
6.1 Quantitative Comparison

Table 3: Simplified sensitivity of global response to high-tie prestress

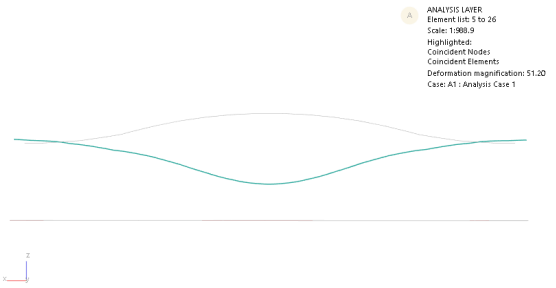
Response Quantity	Prestress Neglected	Prestress Included	Change With Prestress
Maximum vertical displacement (mm)	490	175.3	−64.2%
Centre rafter bending moment (kNm)	9811	198.2	−98.0%
Tusk-to-arch connection bending moment (kNm)	16490	5837	−64.6%
High tie force (kN)	4296	9320	+116.9%
Horizontal reaction at supports (kN)	4211	3480	−17.4%



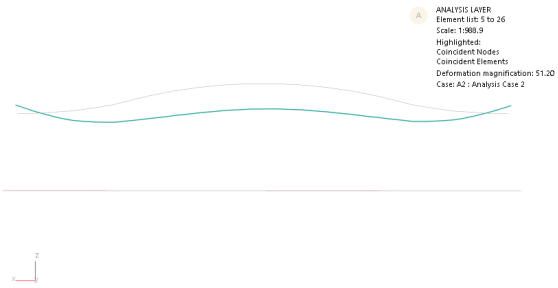
Bending Moment: Prestress Neglected



Bending Moment: Prestress Included



Deformed Shape: Prestress Neglected



Deformed Shape: Prestress Included

Figure 5: Comparison of bending-moment and deformed-shape response with and without high-tie prestress.

6.2 Discussion of Results

Table 3 and Figure 5 show that prestress changes the global response markedly. Without prestress, the central arch sags under gravity and the two tusks rotate upward to maintain continuity at the arch-tusk junctions. With prestress, the central arch retains a clearer curved profile, so the deformation pattern reverses and the tusks are pushed downward. When the arch sags more, the roof behaves in a more flexural manner and the moment distribution can change sign over parts of the frame, whereas the prestressed case remains closer to arch action.

6.3 Implications for Design and Tolerance

These results show that prestress is critical to design interpretation and tolerance assessment. If prestress is neglected, the model overestimates deflection and rafter bending but underestimates the tensile demand in the high tie and its anchorage. This is consistent with the supplementary construction sequence, where staged stressing was used to control geometry and force state, so the prestressed model is adopted as the minimum realistic representation of the completed roof. As shown in Table 2, the change with prestress is significant would imply that without applied prestres, the arch would not be within construction tolerance. Therefore the arch should be installed after prestress is applied.

7 Deliverable 6: Boundary Conditions for More Detailed Analysis

For a more detailed FE study, the rafter splice between the tusk rafter and arch rafter should be extracted from the global frame model and analysed as a local submodel. The purpose of the local model would be to resolve the stress distribution in the splice plates, bolts or welds, and the adjacent box and I-section walls, which cannot be captured accurately by the beam idealisation used in the global model.

The local model should include a short length of member on each side of the splice so that the boundary conditions are applied away from the connection itself. This reduces artificial stress concentrations caused by restraining the model too close to the detail. A practical choice would be to include approximately one section depth of the arch rafter and one section depth of the tusk rafter beyond the splice on either side.

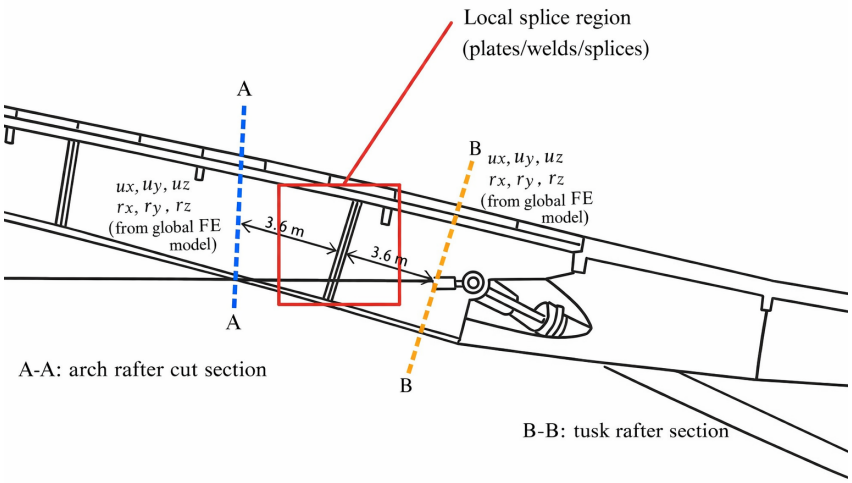


Figure 6: Recommended boundary conditions for the local splice model

The preferred boundary condition is to impose the displacement and rotation components extracted from the global beam model at the two cut faces of the local model. In practice, each cut face can be coupled to a reference node so that the sectional deformation remains compatible with the global model. The six kinematic quantities imposed at each end are the three translations (u_x, u_y, u_z) and the three rotations (r_x, r_y, r_z). This approach is preferable to simply fixing one end, because it reproduces the actual deformation state of the connection and avoids introducing artificial restraint moments.

The corresponding section forces should also be checked against the global model outputs. The main actions expected at the splice are axial force N, vertical shear V, bending moment M, and a smaller torsional component due to the curved roof geometry and eccentricity of the connection detail. If the software or modelling approach makes direct displacement transfer difficult, an acceptable alternative is to apply the equivalent set of sectional resultants at the two cut faces, provided that overall equilibrium is maintained.

To prevent rigid body motion in the local model, one reference node may be lightly stabilised after the global displacements have been assigned. This stabilisation should only remove numerical singularities and must not alter the transferred load path. Contact, bolt preload, or weld representation can then be added depending on the intended level of detail. For the purposes of this coursework, the key requirement is that the splice boundaries are driven by the global model response rather than by arbitrary fixed supports.

Section cuts are located approximately one local section depth away from the splice connection along the axis of the member. This avoids local stress disturbance. As the depth varies along the rafter, the local section depth is equal to the splice-region depth (3.6 m), cuts A–A and B–B are placed approximately 3.6 m from the splice on either side. At both cut faces, the translational and rotational displacements extracted from the global FE model are imposed. This ensures compatibility between the local splice model and the global structural response.

References

[1] British Standards Institution. BS EN 1991-1-3:2003+A1:2015 Eurocode 1: Actions on structures. General actions. Snow loads, 2015.

[2] Oasys. GSA Documentation: Elements. `https://docs.oasys-software.com/structural/gsa/references/hidr-data-element/index.html`, 2026. Accessed March 20, 2026.